MySQL Practice

1. Running the program using the MySQL 8.0 Command Line Client

1.1 Enter the command to display all databases.

```
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 20
Server version: 8.0.35 MySQL Community Server - GPL

Copyright (c) 2000, 2023, Oracle and/or its affiliates.

Oracle is a registered trademark of Oracle Corporation and/or its
affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> show databases;
+--------------------+
| Database           |
+--------------------+
| cool_company2      |
| coolcompany        |
| information_schema |
| mysql              |
| performance_schema |
| sys                |
+--------------------+
6 rows in set (0.00 sec)

mysql>
```

1.2 Select the database named "sys".

```
mysql> use sys;
Database changed
mysql>
```

1.3 Show all tables in the database "sys".

```
mysql> show tables;
+-----------------------------------------------+
| Tables_in_sys                                 |
+-----------------------------------------------+
| host_summary                                  |
| host_summary_by_file_io                       |
| host_summary_by_file_io_type                  |
| host_summary_by_stages                        |
| host_summary_by_statement_latency             |
| host_summary_by_statement_type                |
| innodb_buffer_stats_by_schema                 |
| innodb_buffer_stats_by_table                  |
| innodb_lock_waits                             |
```

1.4 Show all data in the "host-summary" table

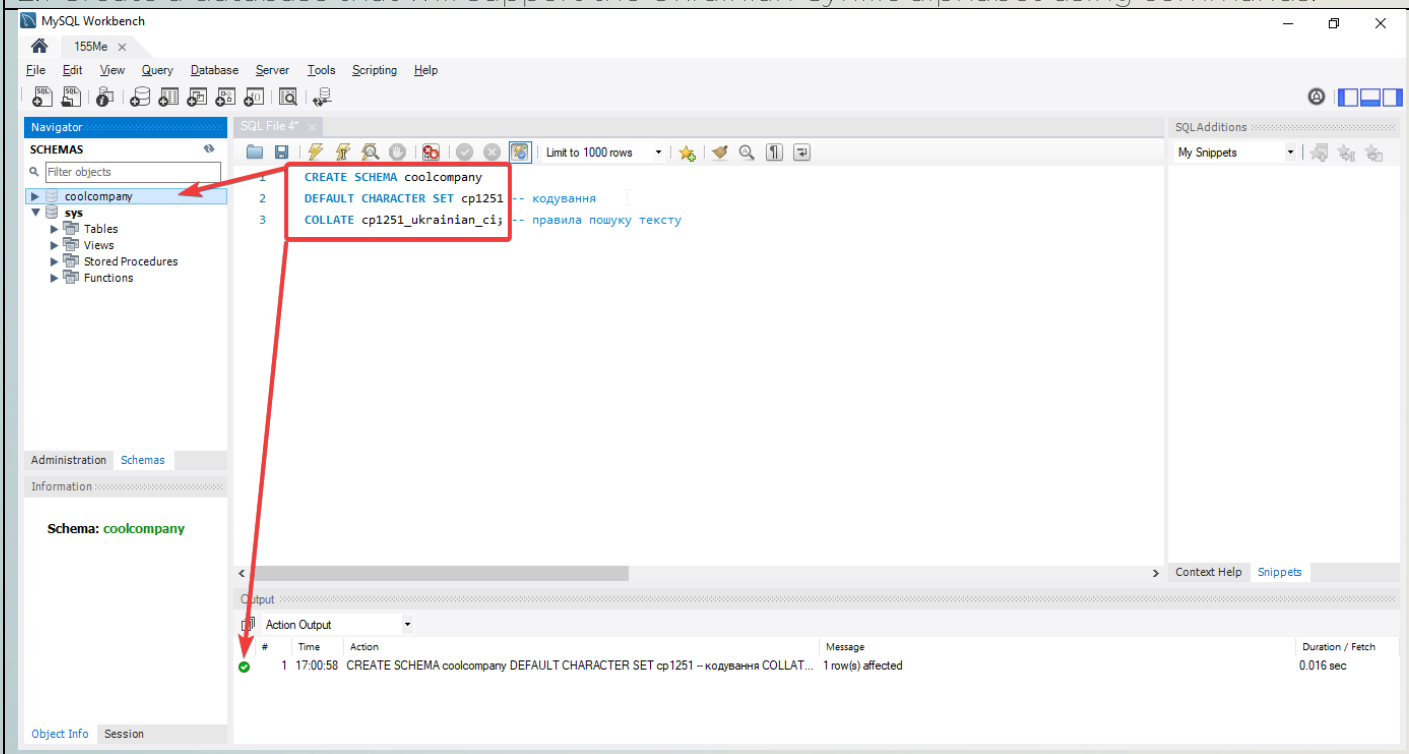
```
mysql> select * from host_summary;
+-----------+------------+-------------------+-----------------------+-------------+----------+-----------------+---------------------+-----------------+
| host      | statements | statement_latency | statement_avg_latency | table_scans | file_ios | file_io_latency | current_connections | current_memory | total_memory_allocated |
+-----------+------------+-------------------+-----------------------+-------------+----------+-----------------+---------------------+----------------+------------------------+
| localhost |       3477 | 7.63 s            | 2.19 ms               |         330 |      279 | 139.24 ms       |                   2 | 6.22 MiB       | 1.25 GiB               |
+-----------+------------+-------------------+-----------------------+-------------+----------+-----------------+---------------------+----------------+------------------------+
1 row in set (0.07 sec)
```

1.5 Exit the command line.

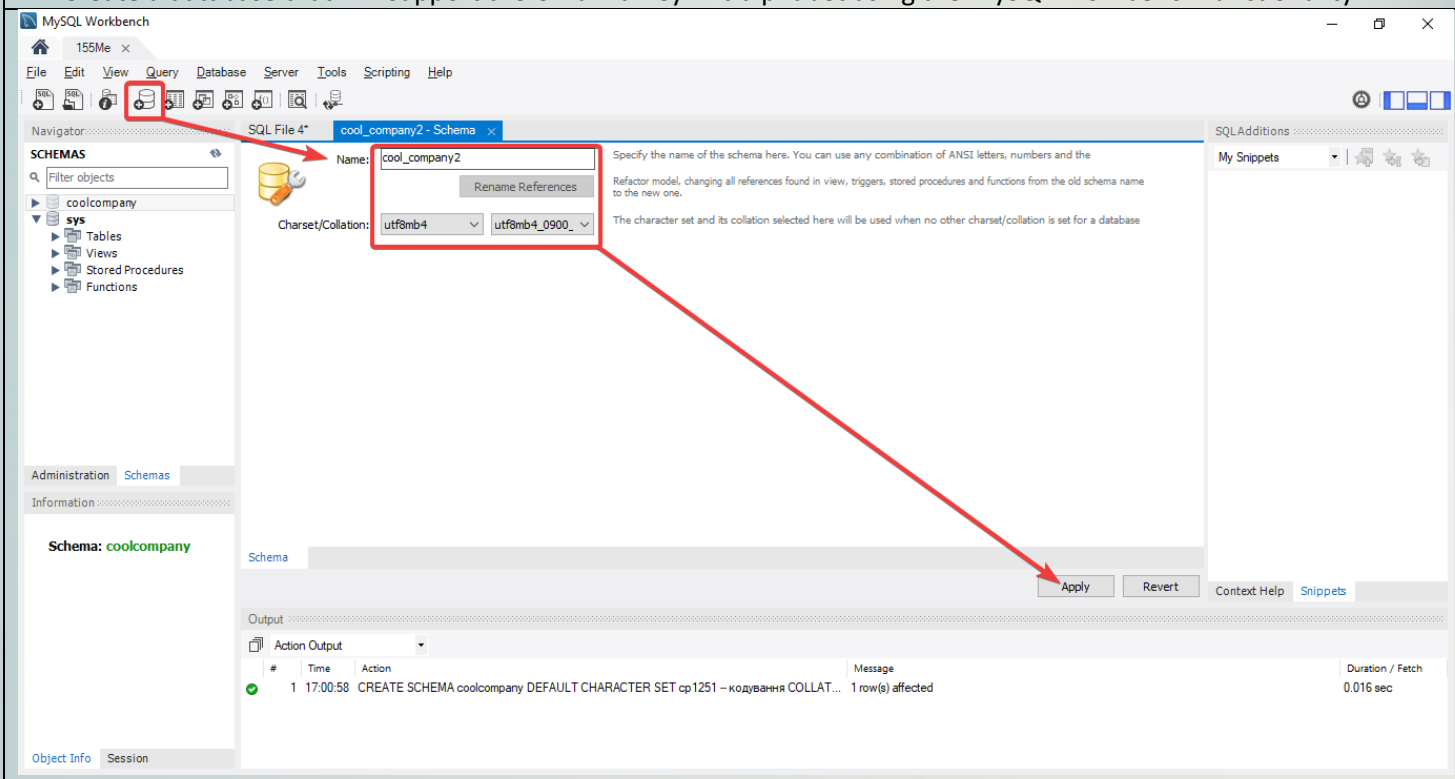
```
mysql> exit
```

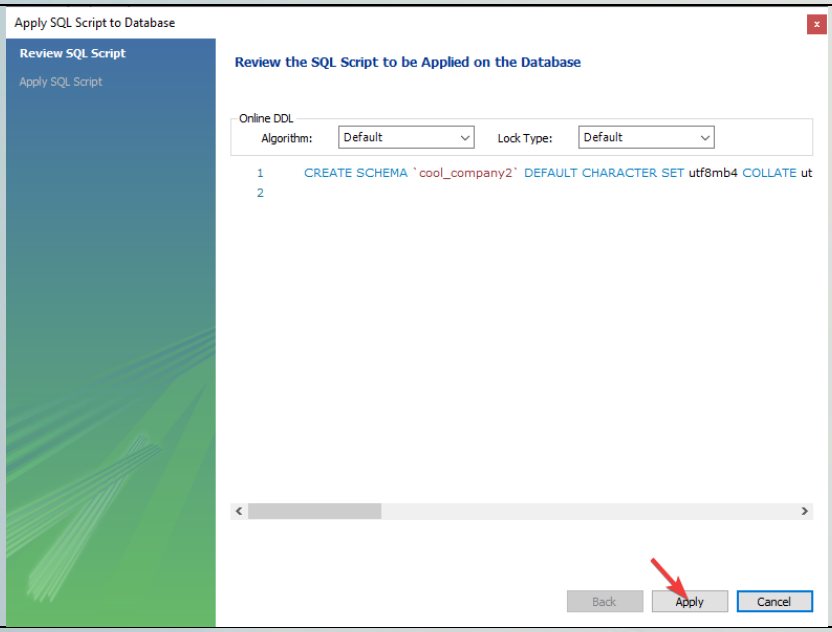
2. Create a database in two ways

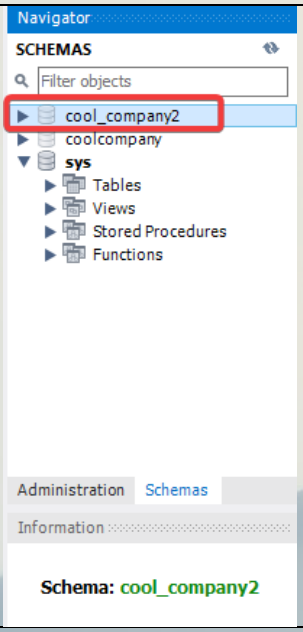
2.1 Create a database that will support the Ukrainian Cyrillic alphabet using commands.



2.2 Create a database that will support the Ukrainian Cyrillic alphabet using the MySQL Workbench functionality.

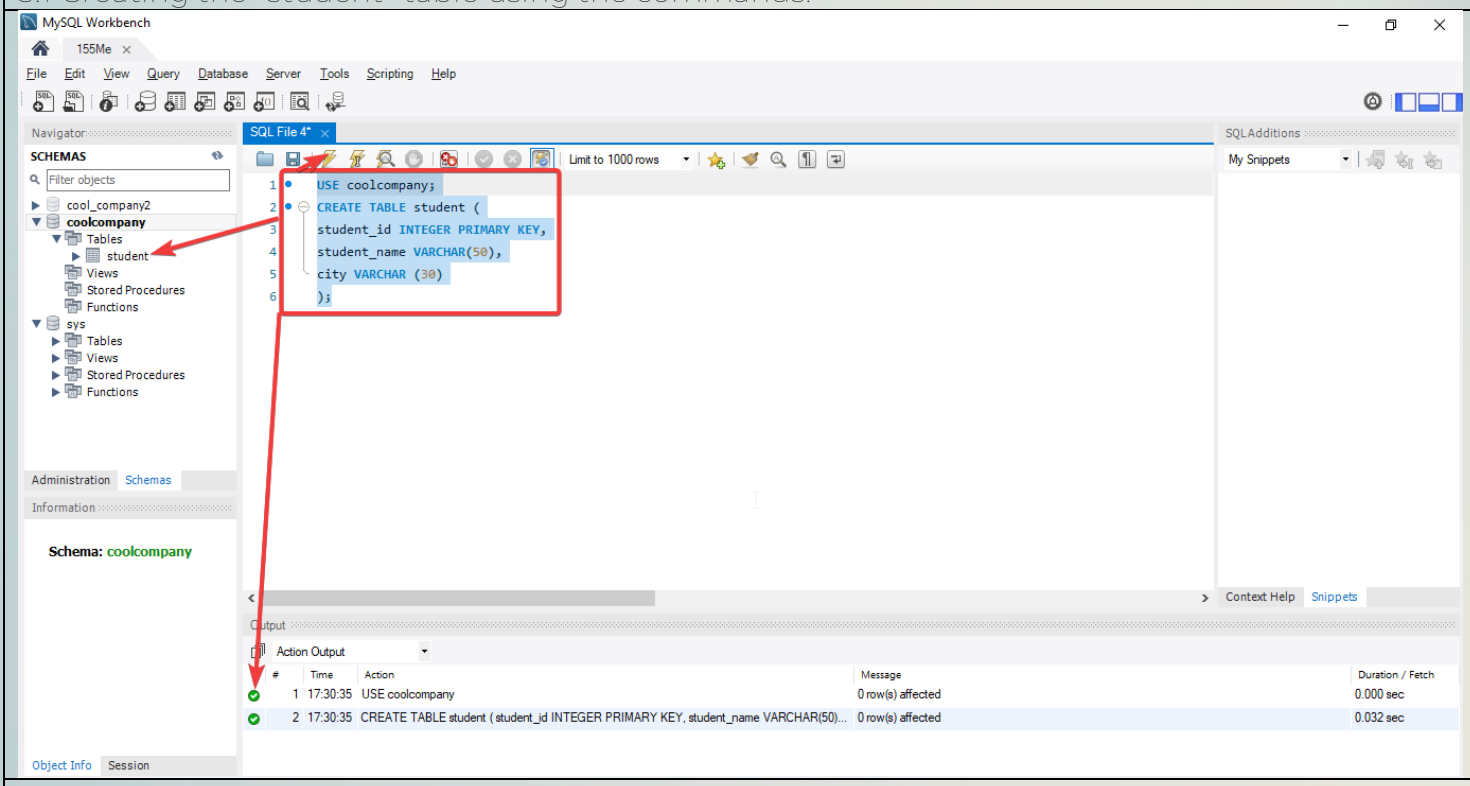




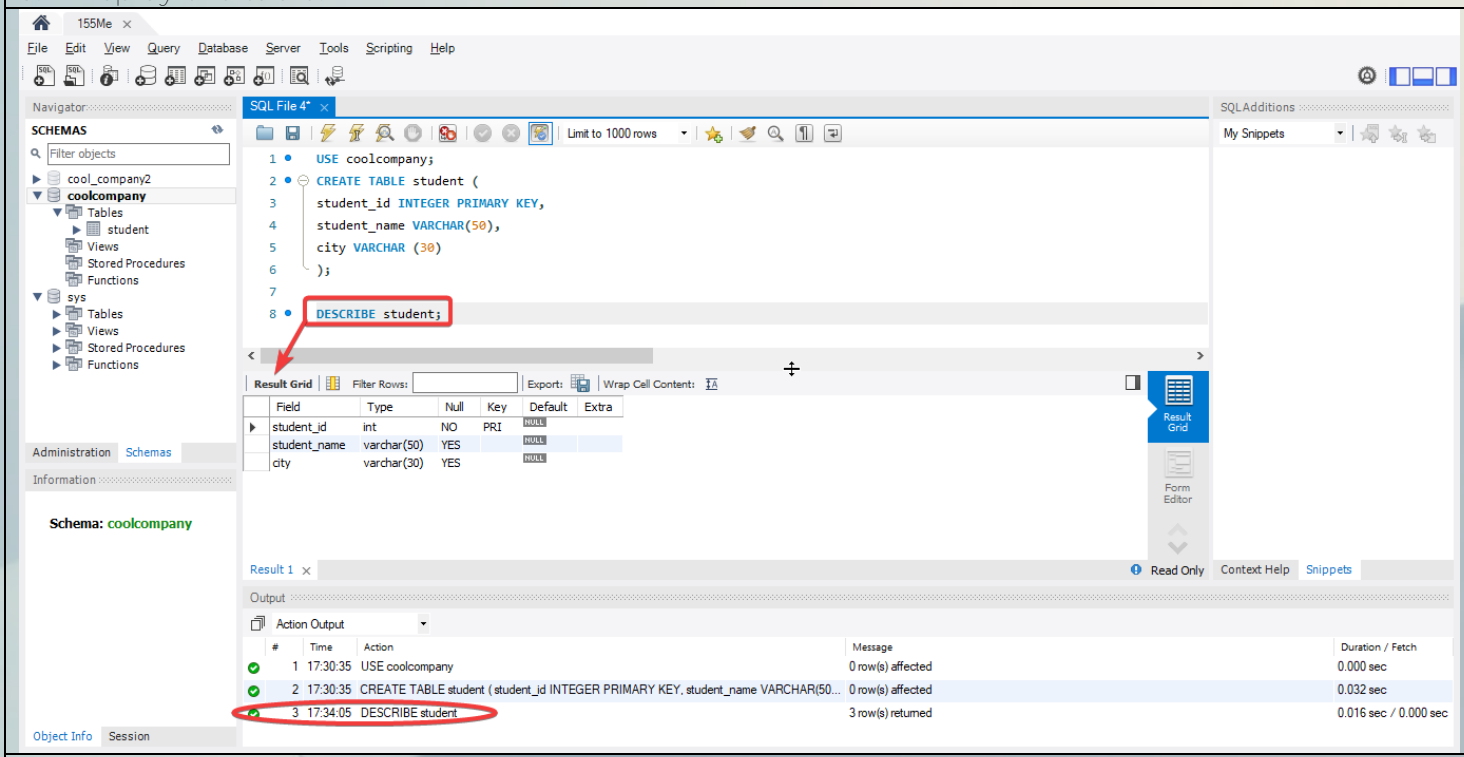


3. How to create a table in the Database (and what is the Primary Key)

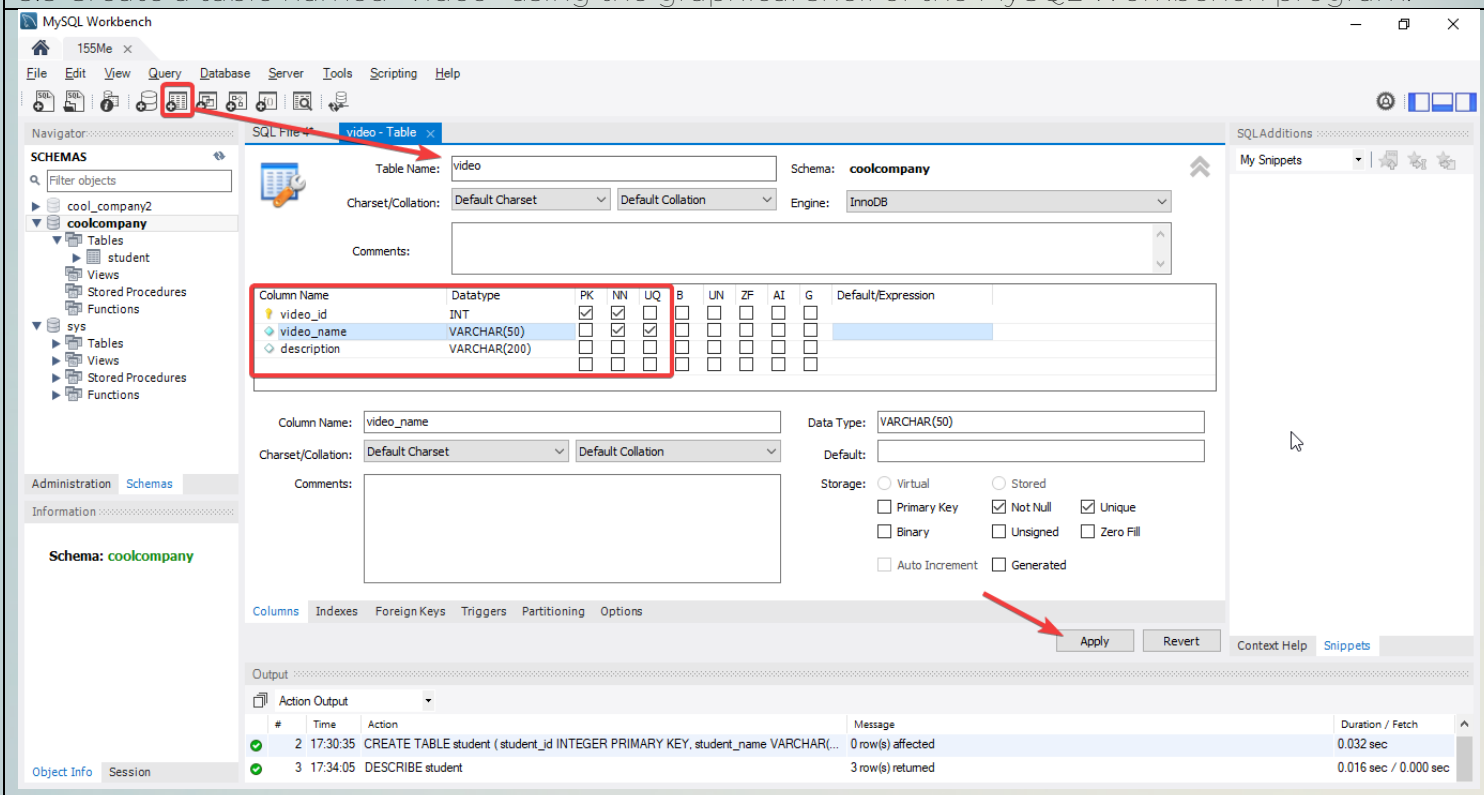
3.1 Creating the "student" table using the commands.

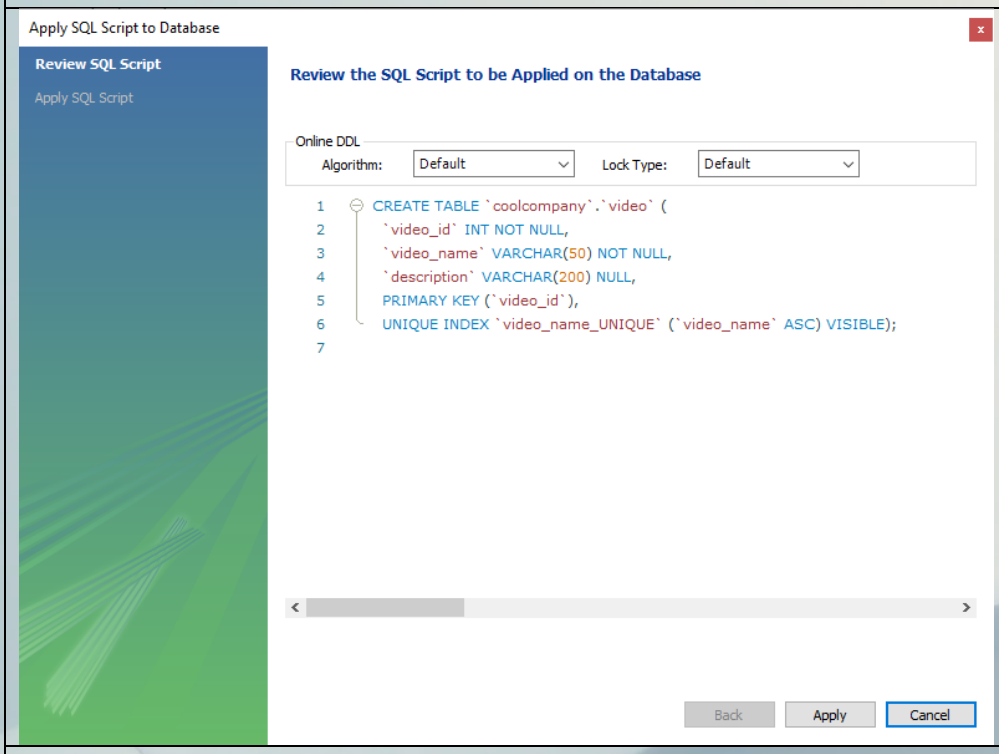


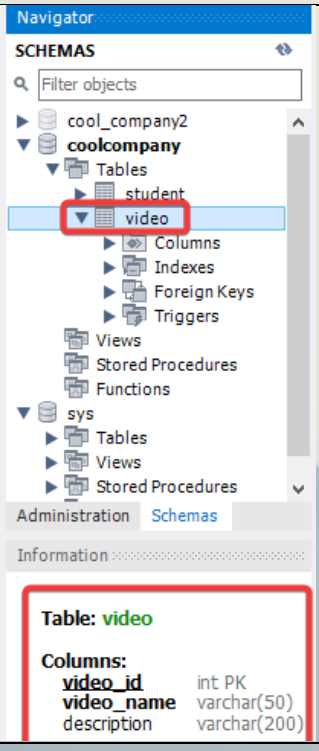
3.2 Display the table.



3.3 Create a table named "video" using the graphical shell of the MySQL Workbench program.







4. How to edit tables in a SQL database

4.1 Add a column with emails to the student table.

155Me

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

cool_company2
coolcompany
Tables
student
Columns
student_id
student_nam
city
Indexes
Foreign Keys
Triggers
video
Views
Stored Procedures
Functions

Administration Schemas

Information

Table: student

Columns:
student_id int PK
student_name varchar(50)
city varchar(30)

Object Info Session

Query 1 SQL File 3*

Limit to 1000 rows

```
1 ALTER TABLE student ADD email VARCHAR(30);
```

SQL Additions

My Snippets

Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	15:23:24	ALTER TABLE student ADD email VARCHAR(30)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.016 sec

155Me

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

cool_company2
coolcompany
Tables
student
Columns
student_id
student_nam
city
Indexes
Foreign Keys
Triggers
video
Views
Stored Procedures
Functions

Administration Schemas

Information

Table: student

Columns:
student_id int PK
student_name varchar(50)
city varchar(30)

Object Info Session

Query 1 SQL File 3*

Limit to 1000 rows

```
1 USE coolcompany;
2 DESCRIBE student;
```

Result Grid Filter Rows: Export: Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
student_id	int	NO	PRI	NULL	
student_name	varchar(50)	YES		NULL	
city	varchar(30)	YES		NULL	
email	varchar(30)	YES		NULL	

Result Grid

Form Editor

Result 2

Read Only

SQL Additions

My Snippets

Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	15:23:24	ALTER TABLE student ADD email VARCHAR(30)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.016 sec
2	15:25:20	DESCRIBE student	4 row(s) returned	0.000 sec / 0.000 sec

4.2 Delete the email column from the Student table.

```
ALTER TABLE student ADD email VARCHAR(30);

ALTER TABLE student DROP COLUMN email;
```

#	Time	Action	Message	Duration / Fetch
1	15:23:24	ALTER TABLE student ADD email VARCHAR(30)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.016 sec
2	15:25:20	DESCRIBE student	4 row(s) returned	0.000 sec / 0.000 sec
3	15:30:04	ALTER TABLE student DROP COLUMN email	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.016 sec

Table: student

Columns:

student_id	int PK
student_name	varchar(50)
city	varchar(30)

```
USE coolcompany;
DESCRIBE student;
```

Field	Type	Null	Key	Default	Extra
student_id	int	NO	PRI	NULL	
student_name	varchar(50)	YES		NULL	
city	varchar(30)	YES		NULL	

#	Time	Action	Message	Duration / Fetch
1	15:23:24	ALTER TABLE student ADD email VARCHAR(30)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.016 sec
2	15:25:20	DESCRIBE student	4 row(s) returned	0.000 sec / 0.000 sec
3	15:30:04	ALTER TABLE student DROP COLUMN email	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.016 sec
4	15:31:29	DESCRIBE student	3 row(s) returned	0.016 sec / 0.000 sec

Table: student

Columns:

student_id	int PK
student_name	varchar(50)
city	varchar(30)

* Attempting to enter more than 30 characters in an email column.

```
INSERT INTO student
VALUES (6, 'Дракон Микола', 'Донецьк', 'MykolaTheGreatDragonOfDonetsk@yahoo.com');
```

#	Time	Action	Message	Duration / Fetch
1	15:46:50	ALTER TABLE student ADD email VARCHAR(30)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.016 sec
2	15:46:55	DESCRIBE student	4 row(s) returned	0.000 sec / 0.000 sec
3	15:47:01	INSERT INTO student VALUES (6, 'Дракон Микола', 'Донецьк', 'MykolaTheGreatDragonOf...	Error Code: 1406. Data too long for column 'email' at row 1	0.000 sec

4.3 Change the size of characters that can be contained in the email column.

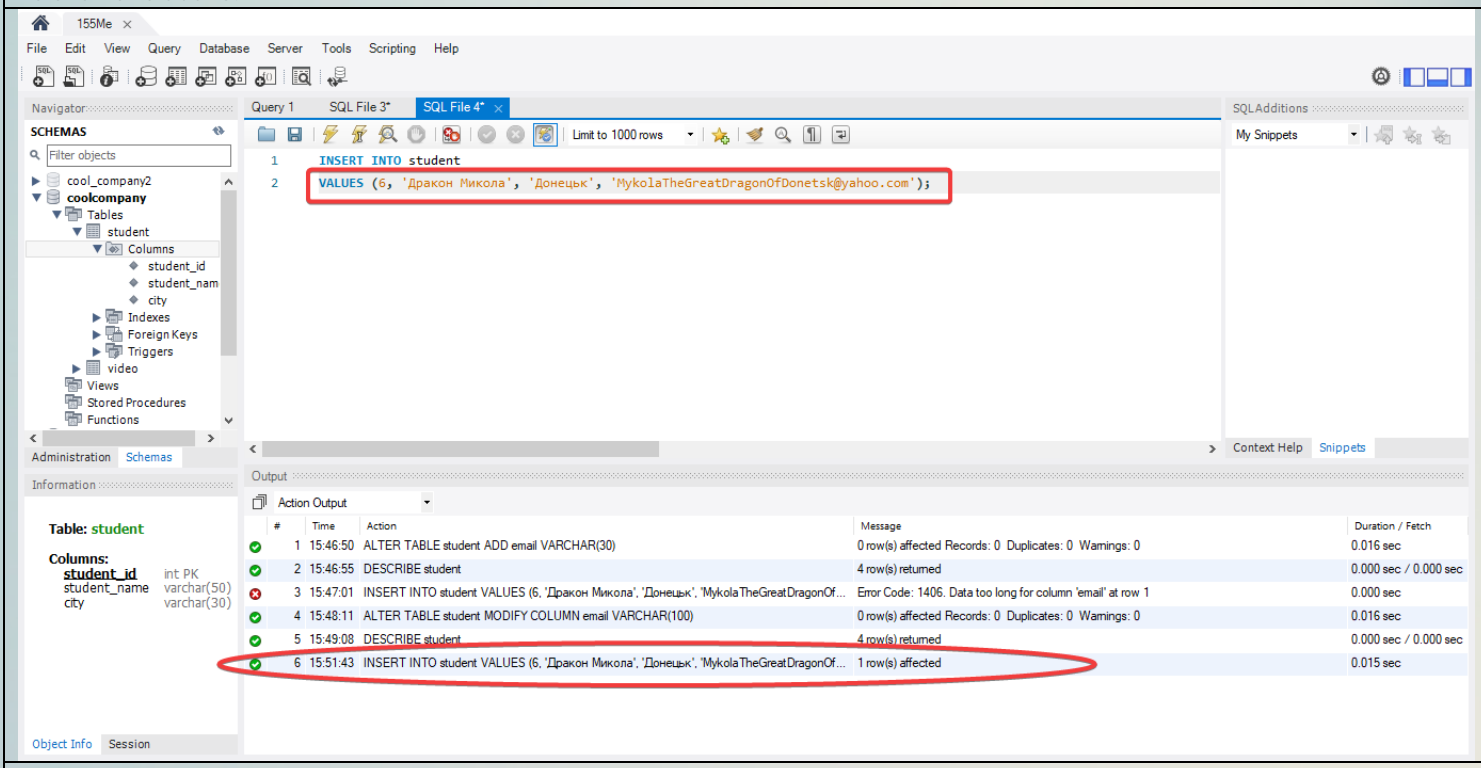
```
ALTER TABLE student ADD email VARCHAR(30);

ALTER TABLE student DROP COLUMN email;

ALTER TABLE student MODIFY COLUMN email VARCHAR(100);
```

#	Time	Action	Message	Duration / Fetch
1	15:46:50	ALTER TABLE student ADD email VARCHAR(30)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.016 sec
2	15:46:55	DESCRIBE student	4 row(s) returned	0.000 sec / 0.000 sec
3	15:47:01	INSERT INTO student VALUES (6, 'Дракон Микола', 'Донецьк', 'MykolaTheGreatDragonOf...	Error Code: 1406. Data too long for column 'email' at row 1	0.000 sec
4	15:48:11	ALTER TABLE student MODIFY COLUMN email VARCHAR(100)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.016 sec

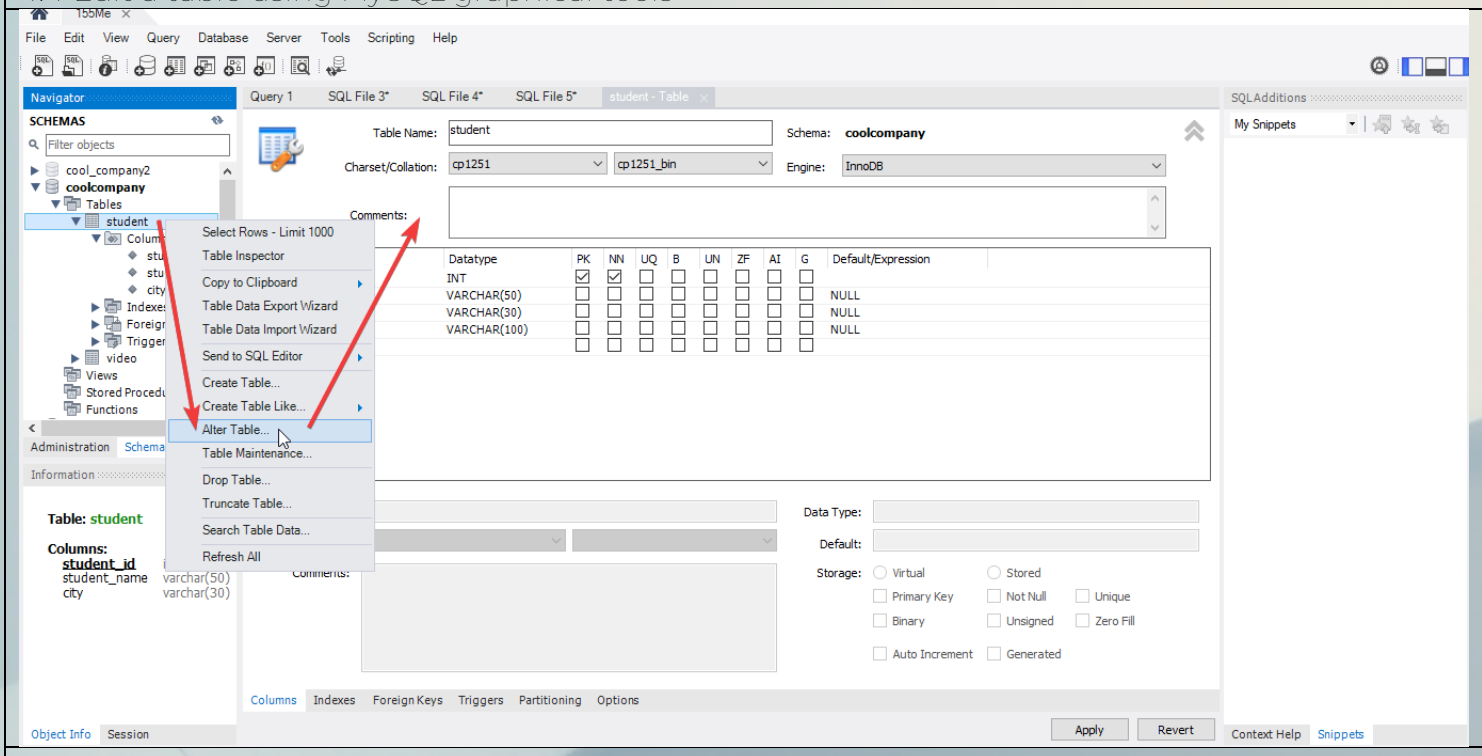
```
USE coolcompany;
DESCRIBE student;
```

Field	Type	Null	Key	Default	Extra
student_id	int	NO	PRI	NULL	
student_name	varchar(50)	YES		NULL	
city	varchar(30)	YES		NULL	
email	varchar(100)	YES		NULL	

** Attempting to enter more than 30 characters in an email column after expanding the column to 100 characters.



4.4 Edit a table using MySQL graphical tools



4.5 Delete the column you just added

Query 1 | SQL File 3* | SQL File 4* | SQL File 5* | student - Table

Table Name: student
Schema: coolcompany
Charset/Collation: cp1251 / cp1251_bin
Engine: InnoDB
Comments:

Column Name	Datatype	PK	NN	UQ	B	UN	ZF	AI	G	Default/Expression
student_id	INT	☑	☑	☐	☐	☐	☐	☐	☐	
student_name	VARCHAR(50)	☐	☐	☐	☐	☐	☐	☐	☐	NULL
city	VARCHAR(30)	☐	☐	☐	☐	☐	☐	☐	☐	NULL
email	VARCHAR(100)	☐	☐	☐	☐	☐	☐	☐	☐	NULL
studentcol		☐	☐	☐	☐	☐	☐	☐	☐	

Move Up
Move Down
Copy
Cut
Paste
Delete Selected
Refresh
Clear Default
Default NULL
Default ''

Column Name:
Charset/Collation: Default Collation
Comments:
Data Type: VARCHAR(45)
Default:
Storage: Virtual Stored
Primary Key Not Null Unique
Binary Unsigned Zero Fill
Auto Increment Generated

Columns | Indexes | Foreign Keys | Triggers | Partitioning | Options

Apply | Revert

4.6 Change the size of characters that can be contained in the email column from 100 to 200.

155Me

File Edit View Query Database Server Tools Scripting Help

Navigator: SCHEMAS
cool_company2
coolcompany
Tables
student
Columns
student_id
student_nam
city
Indexes
Foreign Keys
Triggers
video
Views
Stored Procedures
Functions

Administration | Schemas

Information
Table: student
Columns:
student_id int PK
student_name varchar(50)
city varchar(30)

Object Info | Session

Query 1 | SQL File 3* | SQL File 4* | SQL File 5* | student - Table

Table Name: student
Schema: coolcompany
Charset/Collation: cp1251 / cp1251_bin
Engine: InnoDB
Comments:

Column Name	Datatype	PK	NN	UQ	B	UN	ZF	AI	G	Default/Expression
student_id	INT	☑	☑	☐	☐	☐	☐	☐	☐	
student_name	VARCHAR(50)	☐	☐	☐	☐	☐	☐	☐	☐	NULL
city	VARCHAR(30)	☐	☐	☐	☐	☐	☐	☐	☐	NULL
email	VARCHAR(200)	☐	☐	☐	☐	☐	☐	☐	☐	NULL

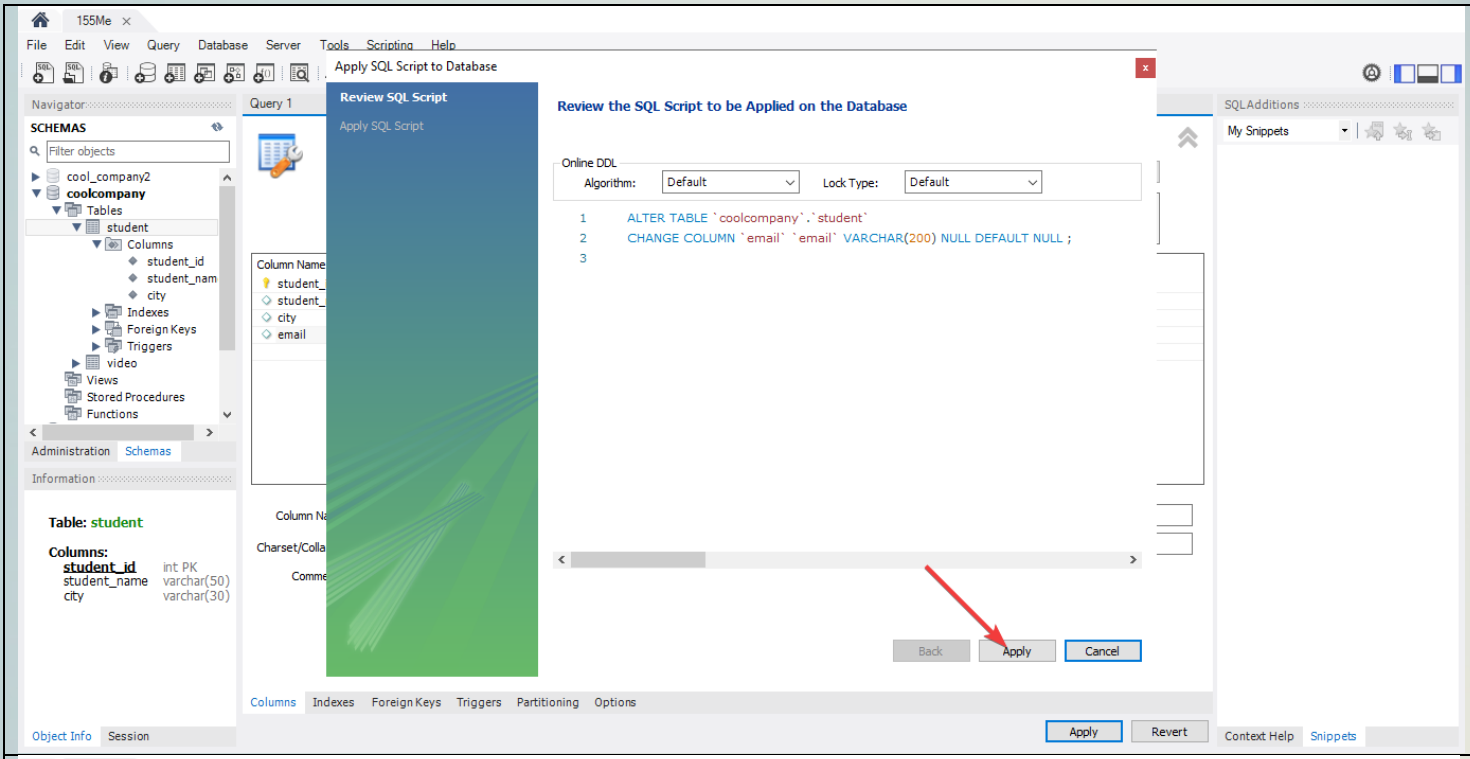
Column Name: email
Charset/Collation: Default Charset / Default Collation
Comments:
Data Type: VARCHAR(200)
Default: NULL
Storage: Virtual Stored
Primary Key Not Null Unique
Binary Unsigned Zero Fill
Auto Increment Generated

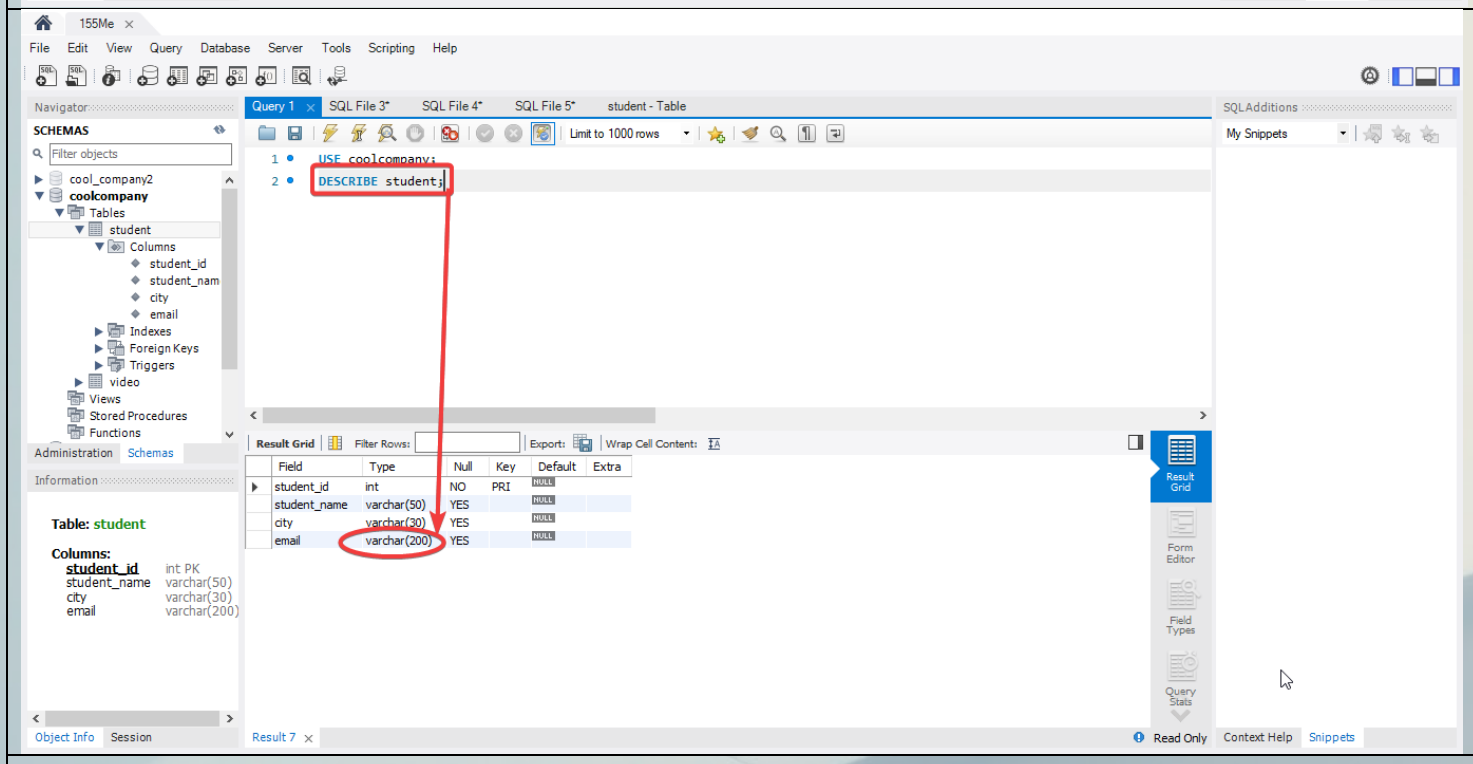
Columns | Indexes | Foreign Keys | Triggers | Partitioning | Options

Apply | Revert

SQL Additions
My Snippets

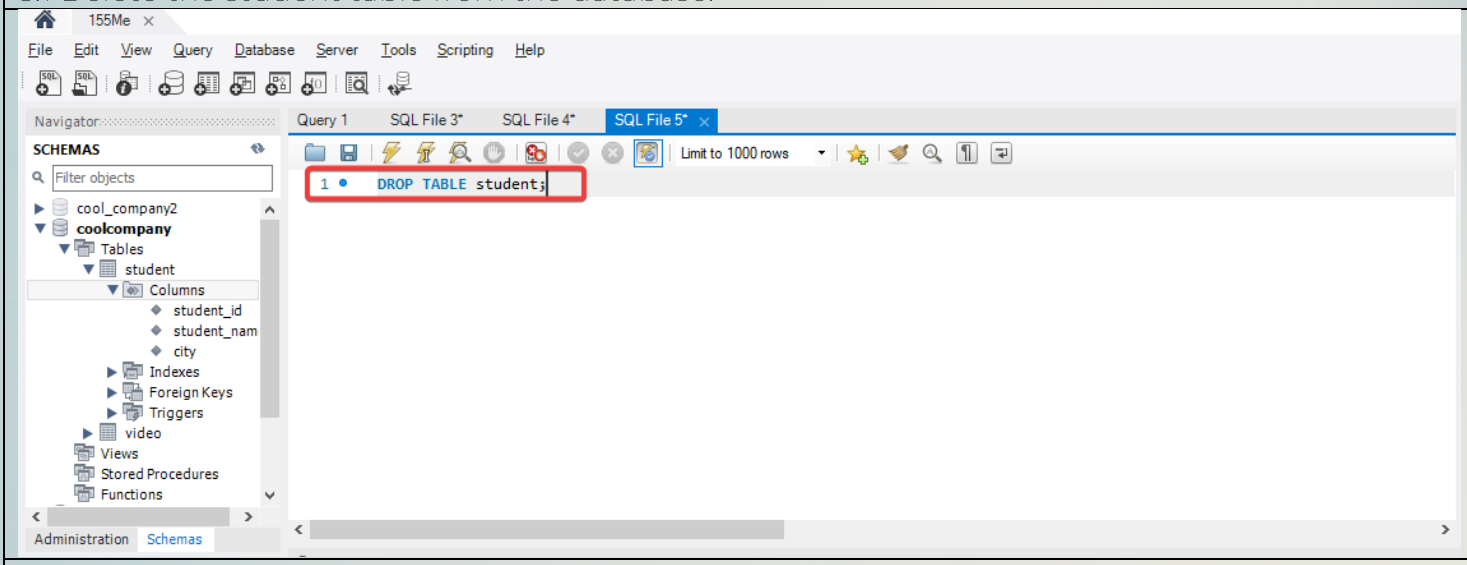
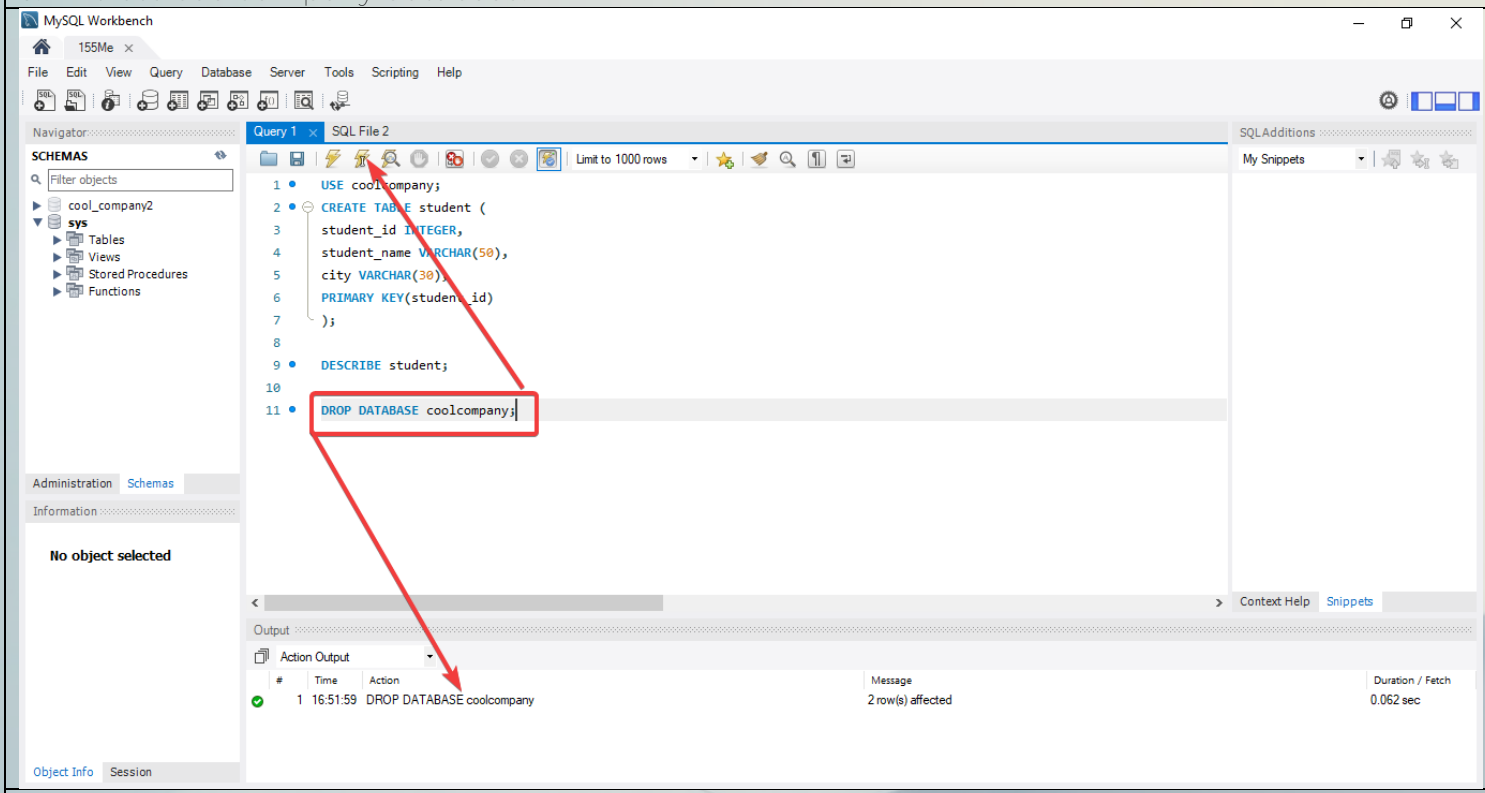
Context Help | Snippets


Apply SQL Script to Database
Review SQL Script
Apply SQL Script
Review the SQL Script to be Applied on the Database
Online DDL
Algorithm: Default
Lock Type: Default
ALTER TABLE `coolcompany`.`student`
CHANGE COLUMN `email` `email` VARCHAR(200) NULL DEFAULT NULL ;
Back
Apply
Cancel
Table: student
Columns:
student_id int PK
student_name varchar(50)
city varchar(30)


USE coolcompany;
DESCRIBE student;
Result Grid
Field Type Null Key Default Extra
student_id int NO PRI NULL
student_name varchar(50) YES NULL
city varchar(30) YES NULL
email varchar(200) YES NULL
Table: student
Columns:
student_id int PK
student_name varchar(50)
city varchar(30)
email varchar(200)

5. Delete a table and database

5.1 Delete the student table from the database.



5.2 Delete coolcompany database.